Racial Segregation and Black-White Longevity Disparities

Thelonious Goerz*

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1 Abstract

Research has documented extensive negative effects of residential, racial segregation on health for Black Americans and mixed results for White Americans. A further layer of uncertainty is whether these associations are causal, which may lead to biased results and mixed findings. In this paper, I use data on early 20th-Century birth cohorts from the 1940 Census and variation from pre-settlement railroads to estimate the causal effect of Black-White segregation on longevity and examine how these effects vary by educational status – as a marker of socioeconomic status. I show that a one percentage point increase in segregation in mid-to-late-life decreases Black longevity by .058 years and increases White longevity by .03 years. In other words, a Black and White individual moving to a place that was 10 percentage points more segregated would live .58 years less and .3 years more, respectively. By educational status, more educated White Americans gain as much as .04 years while the lowest-educated Black Americans face up to a .10 year longevity penalty and the highest educated face no penalty.

*Cornell University

2 Introduction

Black-White racial segregation is a defining feature of American cities both because of its elevated levels across space and time but also because of its exceptionally high levels in comparison to other ethnoracial groups and its significance in maintaining racial inequality (Elbers 2021; Frey and Farley 1996; Massey and Denton 2003). Historically, segregation peaked in the 1960s and 1970s, following the Great Migration of Black residents to Northern Cities starting in the 1950s and earlier in the 1910s (Logan and Stults 2022; Tolnay n.d.). Between 1990-2020, despite Black-White racial segregation declining by 23% overall, average levels still remained at least 10 points, measured by Theil's H index, higher than any other pairwise group comparison (Elbers 2021). Moreover, by 2020, Black populations living in metropolitan areas with greater than 20% Black residents experienced segregation levels between 56-58 dissimilarity points (Logan and Stults 2022:5). These trends and patterns of segregation indicate that, in both contemporary and historical perspective, segregation has remained high and demographically significant.

In the context of persistently high levels of racial segregation, Black-White disparities in health, longevity, and socioeconomic resources also remain wide and significant. For example, Black-White life expectancy gaps vary widely across space and time with one study demonstrating that gaps vary as much as 6-15 years (Harper, MacLehose, and Kaufman 2014). Recently, researchers have highlighted the profound role of structural racism that shapes population health outcomes, socioeconomic status, and the residential environments of Black Americans (Brown and Homan 2024; Collins and Williams 1999; Hendi 2024; Krysan and Crowder 2017). Historical racial inequality, such as Jim Crow (Wrigley-Field 2024), discriminatory housing policy (Rothstein 2018; Taylor 2019), and redlining are prominent pathways affecting Black Americans' health outcomes (Graetz and Esposito 2023) and racially patterned neighborhood sorting processes. Simultaneously, explanations for Black-White health inequalities point to the importance of socioeconomic resources (Link and Phelan 1995;

Phelan and Link 2015), structural processes that channel Black individuals into disadvantaged economic trajectories (Hayward et al. 2000), and place-based factors such as segregation (Collins and Williams 1999; Hendi 2024; Vu, Green, and Swan 2024). However, structural racism and socioeconomic status both jointly shape residential decisions and individual health outcomes. Because of the multiple, interlocking processes that shape residence and health, associations between segregation and longevity cannot be interpreted as causal therefore making it impossible to disentangle whether the effects of segregation on health actually reflect the effects of the residential environment or economic or structural racism pathways (Hendi 2024).

The goal of this paper is to understand whether the effects of racial segregation on longevity are causal and whether they vary by socioeconomic status differences for Black and White Americans. To examine the effect of segregation on longevity, I use data from the 1940 Census linked to CenSoc NUMIDENT death records and an instrumental variable strategy that uses plausibly exogenous variation from 19th-Century railroad lines as an instrument for racial segregation (Ananat 2011). I study longevity, which is defined as the number of years one lives conditional on surviving to 65 years old, because it is a important aspect of population health, a marker of individual life chances, and a cumulative summary measure health status across the life-course (Weber 1922; Wrigley-Field 2024). Results indicate that segregation decreases Black longevity by 5.8 years and increases White longevity by 2.8 years overall. Yet, the positive effects of segregation increase with greater education for White Americans but for Black Americans, higher levels of education do not. These results emphasize the importance of the effects of structural racism on health beyond what can be explained by socioeconomic differences between groups.

3 Background

3.1 Racial Segregation and Health Outcomes

Because of the exceptional nature of Black-White racial inequalities in health (Hayward et al. 2000; Wrigley-Field 2024), wealth (Derenoncourt 2022), education (Montez and Bisesti 2024), and life expectancy (Hendi 2024), and the historical persistence of segregation, scholars have drawn links between Black-White health inequalities and unequal residential arrangements. Theoretically, the residential separation between Black and White Americans is thought to be a physical manifestation of racial inequality and a driving force behind persistent inequality because of the stark socioeconomic and resource differences between Black and White neighborhoods (Blokland 2008; Sharkey 2013) . Early work by W.E.B. DuBois documented these differences in Philadelphia as early as the 1900s (Du Bois 1899) and contemporary patterns continue to have implications for the concentration of poverty in Black neighborhoods (Massey and Denton 2003) and decreased work and education opportunities (Wilson 1996) which lead to resource disparities exacerbating health inequality (Link and Phelan 1995; Phelan and Link 2015; Rothstein 2018).

More recently, empirical and theoretical research has shown that there are persistent negative effects of segregation on the health outcomes ranging from low-birthweight to self-rated mental health for Black Americans and limited unclear patterns for White Americans (Collins and Williams 1999; Smith 2024; Sudano et al. 2013; Vu et al. 2024). In an omnibus review of studies linking segregation to health, Yang, Park, and Matthews (2020) show that negative associations between segregation and health persist across measures and populations of interest for Black Americans, however the majority of studies do not make causal claims or rely on small sample sizes. Despite these limitations of prior work, three notable papers have investigated the causal effects of segregation on health outcomes in compelling ways. Using an instrumental variable strategy introduced by Ananat (2011) leveraging exogenous variation in pre-settlement railroads, Austin, Harper, and Strumpf (2016) found that segregation decreased the birthweight of Black children by 2.8 grams and decreased the birthweight of White children by .63 grams. Similarly, Vu et al. (2024) also study effects on birthweight using the same instrument and find negative impacts of segregation on the birthweight of Black children but find no effects for non-Hispanic White children. Their results indicate that the primary mechanisms that drive these outcomes arise from disparities in access to prenatal care for Black women, structural racism, and food insecurity. Finally, the study that this paper is most closely related to is Hendi (2024) that shows that segregation is responsible for a 16 year increase in the Black-White life expectancy gap for men and a 5 year increase for women, based on period life expectancy estimates between 1990-2019. Nevertheless, this study makes important contributions to the empirical by exploring the effects of segregation on individual-level longevity that facilitate the understanding of how segregation affects health cumulatively and across the socioeconomic gradient.

While studies have documenting the effects of segregation on Black American's health and economic outcomes has been extensively documented, important methodological questions remain about whether these associations are causal and substantively how these patterns may vary across socioeconomic status for Black and White Americans. Research on the persistence of segregation over time draws attention to the complex ways in which individual preferences for neighborhood racial composition and structural sorting patterns affect where people live (Bruch and Mare 2006; Krysan and Crowder 2017). For instance, it could be that patterns of residential segregation and individual health and longevity are driven due to unobserved heterogeneity, preferences, or neighborhood characteristics like poverty and environmental externalities which result in the segregation-health relationship. Moreover, research shows that racial disparities in health persist at higher economic levels (Boen 2016; Esposito 2019) and that more advantaged Black populations often live in similarly segregated neighborhoods to poorer populations (Pattillo-McCoy 2000). Therefore, examining how effects vary by economic status contributes clarity about whether negative health outcomes for Black Americans are primarily the result of differences economic positions or the ability to convert resources to health for Black Americans or wherther disparities are driven by structural racial inequality that transcends socioeconomic status.

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Residential, racial segregation remains a distinctive feature of American cities because of its historically elevated levels and modest declines since the passage of the Civil Rights Act of 1968 (Bader and Warkentien 2016; Elbers 2021; Logan and Stults 2022). Despite the end of legal segregation, between 1990, Black-White segregation declined only 23 percent nationally (Elbers 2021: 2) to an average level just above 30 percent. Black-White segregation levels are highly distinct from those of Latinx and Asian groups because levels are consistently the highest. Moreover, while national trends show a decline, earlier work by Logan (2013) shows that patterns of racial segregation are uneven across places and metropolitan areas are consistently elevated compared to other locations.

This stagnation in the decline of segregation is particularly worrying because of the myriad negative health and economic consequences of segregation for Black populations (Hess et al. 2019; Smith 2024; Williams 1999). In both direct and indirect ways, segregation creates concentrated structural and place-based disadvantages for Black residents (Massey and Denton 2003; Rothstein 2018; Sampson and Sharkey 2008; Taylor 2019).

Prior literature has established a relationship between socioeconomic resources and health (Link and Phelan 1995). In short, ultimate health inequalities are driven by socioecoomic resource differences. Defining work Wilson (1987) demonstrated that rising inner city joblessness shaped by segregation and White exit from metros, left low-income Black neighborhoods with high rates of disadvantage, crime, and limited economic prospects. Because segregation results in the racially stratified allocation of material and social resources tied to neighborhoods, it is plausible that segregation structures Black-White health differences in a similar fashion (Faber and Drummond 2024; Hwang and McDaniel 2022; Williams and Collins 2001).

Research has demonstrated specific negative associations between segregation and health outcomes for Black Americans (Smith 2024; Vu et al. 2024; Yang et al. 2020) At a macro-level, segregation has also been linked to persistently wide Black-White life expectancy gaps at the United States and is considered to be a major driver of Black-White health inequalities (Hendi 2024; Massey and Denton 1989, 2003; Williams 1999). However, causal evidence linking segregation to health is limited and many studies rely on correlational analyses which may confound this relationship. As established by numerous studies, racial segregation persists in part due to structural patterns of racial inequality that may affect health and through individual preferences for where Black and White individuals choose to live (Krysan and Crowder 2017). Because both structural racism and the places individuals live are both associated with health in direct and indirect ways, the relationship between segregation and longevity is endogenous (Austin et al. 2016; Deryugina and Molitor 2021; Graetz, Boen, and Esposito 2022; Homan, Brown, and King 2021; Khan et al. 2023). Thus, precise estimates of the sign and direction of longevity from traditional regression models will be biased.

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3.2 Theoretical Background

Structural pathways

In recent years, scholars have noticed that explanations for Black-White health inequality rooted in hypotheses about socioeconomic differences may not fully capture the residual and persistent negative health status of Black Americans that may interact with socioeconomic gradients (Boen 2016; DeAngelis 2022; Esposito 2019). Even as Black Americans ascend the economic ladder, they experience diminished health compared to

White Americans (DeAngelis, Hargrove, and Hummer 2022). Theory on the role of structural racism shaping health inequality fills these gaps by describing how interconnected social, legal, political, and education systems shape racial inequality within institutions and across domains of society (Brown and Homan 2024: 2). To date, the majority of studies linking structural racism to health have focused on the role of micro-level areas [Brown and Homan (2024); 3], however the pathways that affect health outcomes operate in relation to administrative units, such as the county or state, that play key political and legal roles in determining individual outcomes. From the standpoint of studying segregation, this perspective suggests that county-level mechanisms and pathways that affect health and longevity can co-occur with micro-level pathways, but may have distinct affects on population health.

At the county level, there are a variety of theoretically relevant and compelling pathways that may shape individual health outcomes. For instance, counties that are more segregated have are more likely to have decreased funding for public goods and public opinion in these places indicates decreased support for redistributive policies that could boost the economic conditions of working class populations (Chyn, Haggag, and Stuart 2023; McGhee 2021). More generally Chyn and Haggag (2022), Gilens (2000), and McGhee (2021) hypothesize the a codeterminant of these processes is greater racial resentment in more segregated counties. Through this mechanism, greater aversion to public policy that would likely improve the economic conditions of low-income Black Americans and decreased government support would likely have differential impacts on economic and material resources that have the potential to affect low-income populations more broadly, regardless of race.

County-level mechanisms also work through the criminal legal system to affect individual outcomes through institutional and governmental mechanisms that are shaped by historical racial inequality (Brown and Homan 2024; Ray 2023; Williams and Collins 2001). Compared to any other sex-age-race group, Black men and women are at the highest risk of experiencing police violence (Edwards, Lee, and Esposito 2019). Bell (2020) demonstrates that patterns of aggressive policing and punitive criminal justice policy contexts stem from high levels of segregation. In this way policing is tied to racial composition and the differing residential arrangements between White and Black Americans. Exposure to aggressive policing in segregated areas may plausibly affect Black men more than White men through arrests and incarceration. Conversely, Light and Thomas (2019) show that one way that segregation increases racial inequalities in mortality is through White Americans' decreased risk of homocide victimization and an increased risk for Black Americans. Even with less extreme outcomes, exposure to aggressive policing leading to a higher likelihood of criminal justice contact has been linked to negative population health outcomes for Black Americans (Cao, Esposito, and Lee 2025; Lee 2024).

Place-based effects of segregation are also shaped by historical processes and long-run patterns of racism and stratification (Graetz and Esposito 2023; Rothstein 2018). Ananat (2011) demonstrates that, in Northern metropolitan areas, segregation led to an increase in poverty for Black Americans and decreased poverty for White Americans. Additionally, segregation increased Black-White income gaps. In contemporary contexts, Graetz and Esposito (2023) show that HOLC redlining maps, which were developed for the sole purpose of maintaining segregation, created greater economic isolation and devaluation of property values in mapped areas that have proximate implications for segregation and health outcomes through homeownershi (Breen 2024b). < Need a conclusion>

Micro-level pathways

At a micro-level, high levels of segregation are associated with processes that lead to disinvestment and disadvantage such as White avoidance, devaluation of Black property values, and an increasing frequency of